

# SARS-CoV-2 NSP13 interacts with TEAD to suppress Hippo-YAP signaling

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## Abstract

## Abstract

The Hippo pathway is critical to organ development, homeostasis, and regeneration, facilitated by YAP/TEAD-mediated gene expression. Although emerging studies report Hippo-YAP dysfunction after viral infection, it is largely unknown in the context of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Here, we analyzed RNA sequencing data from SARS-CoV-2 infected human lung samples and induced pluripotent stem cell-derived cardiomyocytes (iPSC-CMs), and observed a decrease in YAP target gene expression. In screening SARS-CoV-2 nonstructural proteins, we found that nonstructural protein 13 (NSP13), a conserved coronavirus helicase, inhibited YAP transcriptional activity independent of the upstream Hippo kinases LATS1/2. Consistently, introducing NSP13 into cardiomyocytes suppressed active YAP (YAP5SA) *in vivo*. Subsequent investigations on NSP13 mutants indicated that NSP13 helicase activity is crucial for suppressing YAP transactivation. Mechanistically, TEAD4 serves as a platform for recruiting NSP13 and YAP. NSP13 inactivates the YAP/TEAD4 transcription complex through its interacting proteins, such as transcription termination factor 2 (TTF2). These discoveries reveal a novel YAP/TEAD regulatory mechanism orchestrated by TEAD4, which provides molecular insights of Hippo-YAP regulation after SARS-CoV-2 infection.

### eLife assessment

This **important** study reports the molecular function of the SARS-CoV-2 helicase NSP13, which inhibits the transcriptional activity of the YAP/TEAD complex *in vitro* and *in vivo*. The evidence supporting the authors' claims is **solid**, with rigorous cell biological assays and multi-omic studies. This work will be of interest to scientists studying COVID-19 infection and the Hippo-YAP signaling pathway.

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## Introduction

The evolutionarily conserved Hippo-YAP signaling pathway integrates various extracellular signals, including mechanical force, cell adhesion, and nutrient availability, through the protein kinases LATS1/2 (1). As the primary substrate of LATS kinases, YAP is inactivated after phosphorylation via cytoplasmic retention or degradation. The unphosphorylated YAP translocates to the nucleus and forms complexes with transcription factor partners, notably TEAD, to activate target genes (1,2) that are involved in tissue development, homeostasis, and regeneration across multiple organs (2–7). YAP/TEAD transactivation is regulated by intricate molecular mechanisms that govern YAP/TEAD activity, localization, and interaction. Dysregulation of these processes has been implicated in various diseases including cancer (8–11).

Recent studies have shown that YAP acts as a natural inhibitor of innate immunity against viral infection (12,13), but conflicting findings were observed in YAP regulation including expression, degradation, and nucleus localization, after multiple types of viral infection (14). Although these observations shed light on the complex dynamics of YAP during viral responses, the specific mechanisms underlying YAP/TEAD regulation after SARS-CoV-2 infection are poorly understood. SARS-CoV-2, which caused the global COVID-19 pandemic, has well-documented effects on the respiratory, digestive, central nervous, and cardiovascular systems (15–17). Studying the effects of SARS-CoV-2 on Hippo-YAP signaling provides insight into the molecular underpinnings of virus-induced pathophysiology. Two recent studies showed opposite YAP activity after SARS-CoV-2 infection (18,19), but the lack of YAP/TEAD target gene expression data prevented a better understanding of these disparate results. Thus, the precise understanding of YAP/TEAD regulation after SARS-CoV-2 infection remains elusive.

Here, we found that SARS-CoV-2 infection reduced YAP target gene expression in lung epithelial cells of patients with COVID-19 and human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs). Using a comprehensive screening of SARS-CoV-2 nonstructural proteins, we identified NSP13 as a key factor in inhibiting YAP transcriptional activity and suppressing active YAP (YAP5SA) activity *in vivo*. Moreover, NSP13 suppression on YAP was dependent on its helicase activity. Mechanistically, NSP13 directly bound to TEAD4 and inhibited the transcriptional activity of the YAP/TEAD4 complex by recruiting transcriptional suppressors such as TTF2. These findings indicate a novel function of NSP13 in regulating YAP/TEAD activity and provide key insights into how the SARS-CoV-2 genome modulates transcriptional activity of the YAP-TEAD complex.

## Results

### SARS-CoV-2 infection suppresses YAP activity in host cells

To assess alterations in YAP activity with SARS-CoV-2 infection *in vivo*, we integrated and reanalyzed single nuclei RNA (snRNA) sequencing data derived from human lung samples (20,21), identifying 10 major cell types (Figure 1A, figure supplement 1A). The expression levels of *TMPRSS2* and *ACE2*, two key entry factors for SARS-CoV-2 infection (22), was higher in lung epithelial cells than in other cell types (Figure 1B, figure supplement 1B). Lung epithelial cells were categorized into alveolar type 1 (AT1) and alveolar type 2 (AT2) (figure supplement 1C, D). SARS-CoV-2 is more likely to infect AT1 cells because they cover >95% of the alveolar surface. The YAP score, evaluated by using 38 YAP target genes expression, was lower in AT1 cells from COVID-19 patients than from controls (Figure 1C). The signature genes associated with AT1 cells including *AGER* and *CLIC5*, reported as YAP target genes (23,24), were downregulated in lung samples from COVID-19 patients (figure supplement 1E). To substantiate this phenotype, we

evaluated a bulk RNA-sequencing dataset derived from hiPSC-CMs exposed to different SARS-CoV-2 concentrations (25) (Figure 1D). The average expression of cardiomyocyte-specific YAP target genes (26) in the dataset revealed that YAP targets were decreased in a dose-dependent manner after SARS-CoV-2 infection (Figure 1E). Several canonical YAP targets in cardiomyocytes are shown in Figure 1F. These data support that SARS-CoV-2 infection suppresses YAP activity in host cells.

## NSP13 inhibits YAP5SA transactivation *in vitro* and *in vivo*

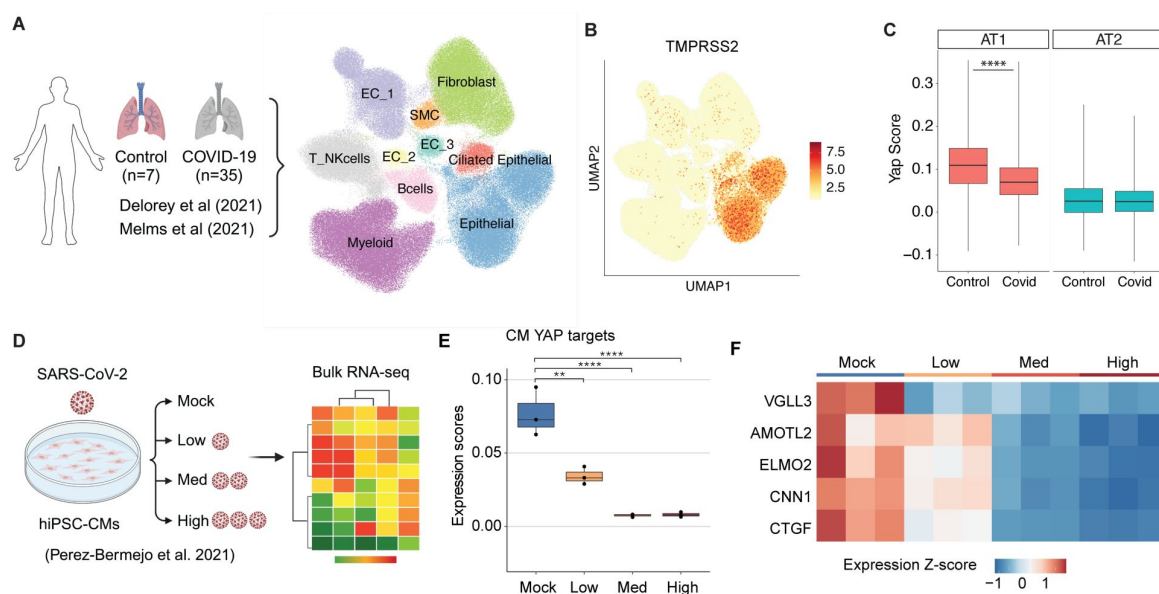
To identify which SARS-CoV-2 protein suppresses YAP activity, we screened 11 NSP proteins by using a dual-luciferase reporter assay. Compared with other NSPs, NSP13 strongly inhibited YAP transcription activity (Figure 2A). By using constitutively active YAP (YAP5SA), which is resistant to phosphorylation and inactivation by LATS1/2 (27), we found that NSP13 suppressed YAP transactivation independent of its upstream kinase LATS2 (figure supplement 2A). Moreover, NSP13 attenuated YAP5SA activity in a dose-dependent manner (Figure 2B).

To investigate NSP13 function *in vivo*, we used YAP5SA transgenic mice (aMyHC-MerCreMer;YAP5SA), which express YAP5SA in cardiomyocytes after tamoxifen administration. This induction led to cardiomyocyte hyperplasia and increased ejection fraction (EF) and fractional shortening (FS), which ultimately resulted in mortality in mice (26). The adenovirus expressing NSP13 (AAV9-NSP13), which specifically infects cardiomyocytes, was used in *in vivo* experiments (figure supplement 2B and Figure 2C). Expectedly, NSP13 expression in YAP5SA mouse cardiomyocytes significantly increased survival rates (Figure 2D) and restored cardiac function, as evidenced by EF and FS (Figure 2E and figure supplement S2C). Furthermore, NSP13 expression reversed the smaller left ventricle (LV) chamber in YAP5SA mouse hearts (26) (Figure 2F, G). To further investigate NSP13 effects on the mouse heart, we collected heart tissue from the surviving mice 21 days after tamoxifen injection. Notably, NSP13 expression in YAP5SA mice reversed heart overgrowth (Figure 2H, I and figure supplement 2D). Overall, these data indicate that NSP13 suppresses YAP activity *in vitro* and *in vivo*.

## NSP13 helicase activity is required for YAP suppression

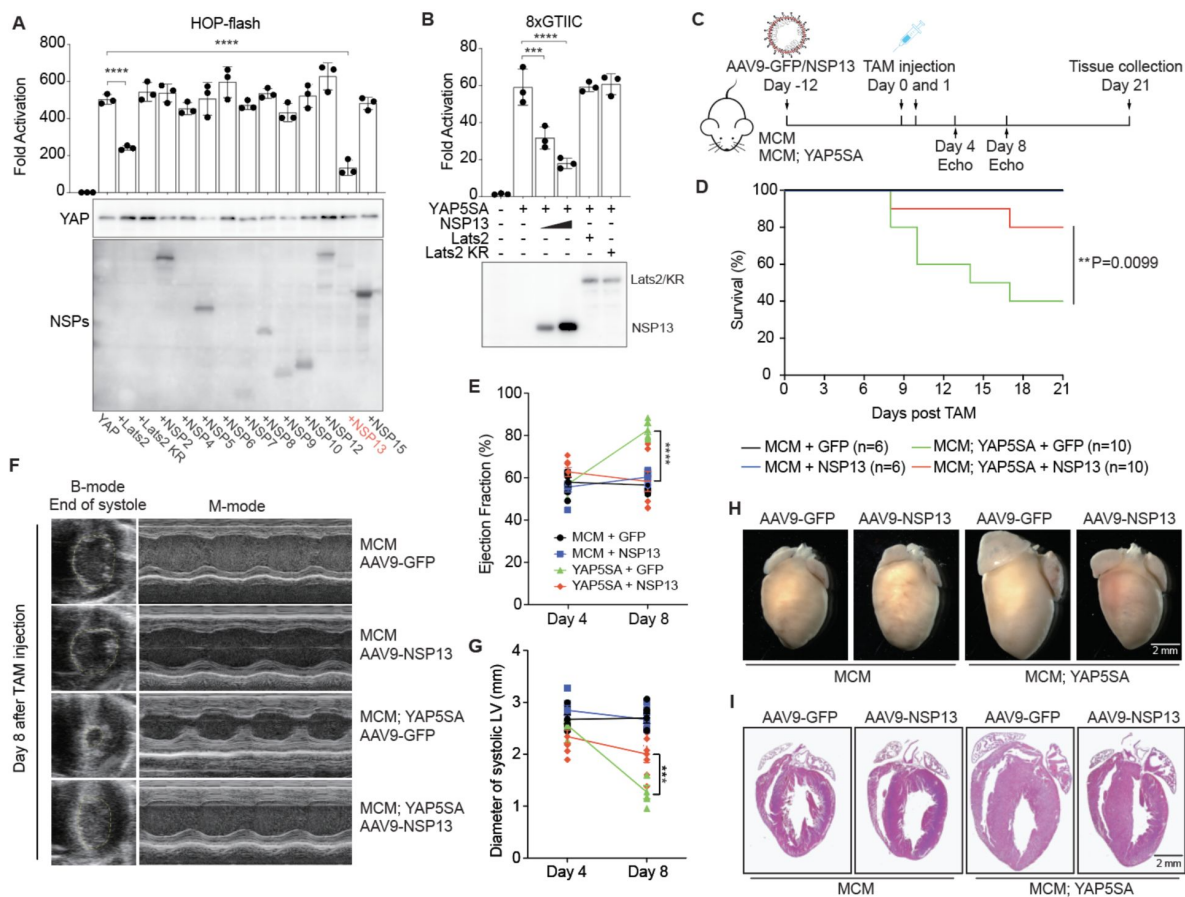
NSP13, a helicase with conserved sequence across all coronaviruses (Figure 3A), plays a critical role in viral replication and is a promising target for antiviral treatment (28–35). K131, K345/K347, and R567 were identified as important amino acid sites for NSP13 helicase activity in SARS-CoV (36). Because of the 99.8% sequence identity of NSP13 between SARS-CoV-2 and SARS-CoV, we believed these sites were similarly crucial for SARS-CoV-2 NSP13. Reporter assays showed that NSP13-K131A with reduced helicase activity was able to suppress YAP, whereas NSP13-R567A (lost ATP consumption) and NSP13-K345A/K347A (obstructed the nucleic acid binding channel) failed to inhibit YAP activity (Figure 3B). To assess the contribution of each domain to YAP regulation, we constructed 6 NSP13 truncations (Figure 3C), but none led to a reduction in YAP transactivation (Figure 3D). These findings suggest that the full length NSP13 with helicase activity is required for to suppress YAP transactivation.

SARS-CoV-2 has consistently mutated over time, resulting in variants that differ from the original virus. We also examined the effect of the NSP13 mutations (31) (Figure 3E and figure supplement 3A) on YAP and found that all mutants can suppress YAP transactivation (Figure 3F and figure supplement 3A).



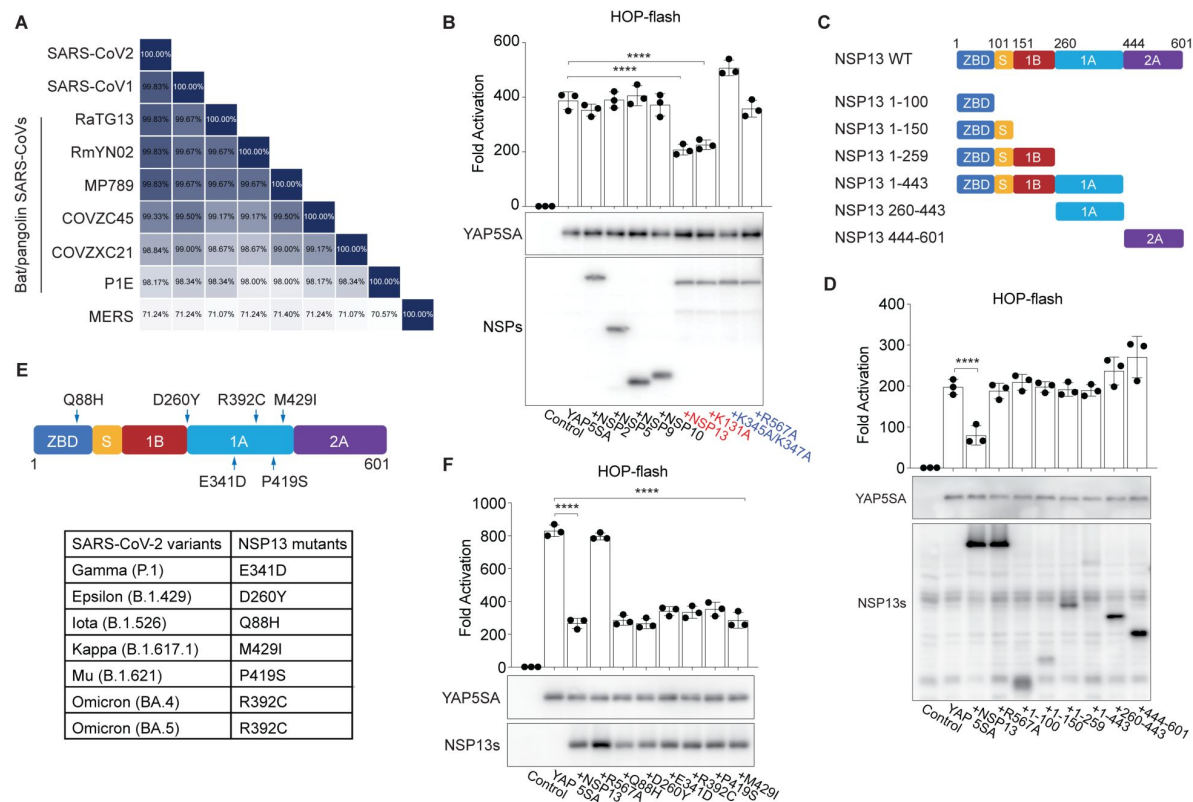
**Figure 1.**

SARS-CoV-2 infection suppresses YAP activity *in vivo* and *in vitro*. **(A)** Overview of integrated single-nucleus RNA sequencing and uniform manifold approximation and projection (UMAP) of cell types in lung samples from controls and patients with COVID-19. EC, endothelial cells; NK, natural killer cells; SMC, smooth muscle cells. **(B)** UMAP visualization of TMPRSS2 expression in the 10 cell types. **(C)** Yap scores in alveolar type 1 (AT1) and alveolar type 2 (AT2) epithelial cells from lung samples in controls and patients with COVID-19. Wilcoxon test, \*\*\*\* $p < 0.0001$ . **(D)** Overview of SARS-CoV-2 infection in human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs). **(E)** Box plot showing the mean expression scores of known YAP target genes in hiPSC-CM bulk RNA-sequencing data. Each dot represents a biological replicate. Student  $t$ -test; \*\* $p < 0.01$ , \*\*\*\* $p < 0.0001$ . **(F)** Heatmap displaying the expression Z-scores of example YAP targets in the iPSC-CM bulk RNA-sequencing data. Each column corresponds to a single biological sample.



**Figure 2.**

NSP13 inhibits YAP transactivation *in vitro* and *in vivo*. **(A)** Screening of 11 NSPs for YAP activation by using a dual-luciferase reporter assay (HOP-flash). Compared with other NSPs, NSP13 strongly inhibited YAP transactivation at low levels of protein expression. (n = 3 independent experiments; data are reported as mean  $\pm$  SD). \*\*\*\*p<0.0001, one-way ANOVA. **(B)** Reporter assay (8xGTIIC) results showing that NSP13 but not YAP upstream kinase LATS2 inhibited YAP5SA transactivation in a dose-dependent manner. (n = 3 independent experiments; data are presented as the mean  $\pm$  SD). \*\*\*p<0.001, \*\*\*\*p<0.0001, one-way ANOVA. **(C)** Experimental design of NSP13 study in mice. Both Control (aMyHC-MerCreMer;WT) and YAP5SA (aMyHC-MerCreMer;YAP 5SA) mice were injected with AAV9-GFP or AAV9-NSP13. At 12 days after virus injection, the mice received two low-dose of TAM (10 ug/g). Cardiac function was recorded by echocardiography at day 4 and 8 after the second shot of tamoxifen. The mouse hearts of all the surviving mice were collected at day 21 post tamoxifen injection. **(D)** NSP13 expression in cardiomyocytes improved survival rate of YAP5SA mice after TAM injection compared to YAP5SA mice with AAV9-GFP infection. \*\*p=0.0099, log-rank (Mantel-Cox) test. **(E)** Ejection fraction in YAP5SA mice was increased on day 8 after tamoxifen injections (10 ug/g x2). NSP13 expression reversed the increase of EF in YAP5SA mice. \*\*\*\*p<0.0001, three-way ANOVA. **(F)** Representative B-mode and M-mode echocardiographic images of mouse hearts in four groups 8 days after tamoxifen (TAM) induction. **(G)** A reduction in the size of the left ventricle was seen in YAP5SA mice at day 8 after tamoxifen injection. NSP13 introduction reversed this trend as evidenced by an increase in the diameter of the left ventricle. \*\*\*p<0.001, three-way ANOVA. **(H-I)** Representative whole mount and hematoxylin and eosin images of mouse hearts at 21 days after tamoxifen induction. Scale bar, 2 mm.



**Figure 3.**

NSP13 helicase activity is required for suppressing YAP activity. **(A)** Conserved amino acid sequences of NSP13 among coronaviruses. **(B)** We constructed SARS-CoV-2 NSP13 mutant plasmids to examine the mechanisms underlying YAP suppression. NSP13-R567A, which loses its ATP consumption ability, did not inhibit YAP5SA transactivation, whereas NSP13 K345A/K347A, which loses its nucleic acid binding activity, mildly promoted YAP5SA transactivation. (n = 3 independent experiments; data are reported as the mean  $\pm$  SD). \*\*p < 0.01, \*\*\*\*p < 0.0001, one-way ANOVA. **(C)** We constructed 6 NSP13 truncations on the basis of the NSP13 domain map. **(D)** Reporter assay results indicated that none of the truncations led to a reduction in YAP transactivation and the NSP13 DNA binding domains 1A and 2A slightly increased YAP5SA activation, suggesting that the full length NSP13 with helicase activity may be required for suppression of YAP transactivation. (n = 3 independent experiments; data are reported as the mean  $\pm$  SD). \*p < 0.05, \*\*\*\*p < 0.0001, one-way ANOVA. **(E)** Summary of NSP13 mutants from SARS-CoV2 variants. **(F)** Reporter assay (HOP-flash) results indicated that NSP13 mutations did not affect its suppression of YAP5SA transactivation. \*\*\*\*p < 0.0001, one-way ANOVA.

## NSP13 interacts with TEAD4 and recruits

### YAP repressors into the YAP/TEAD complex

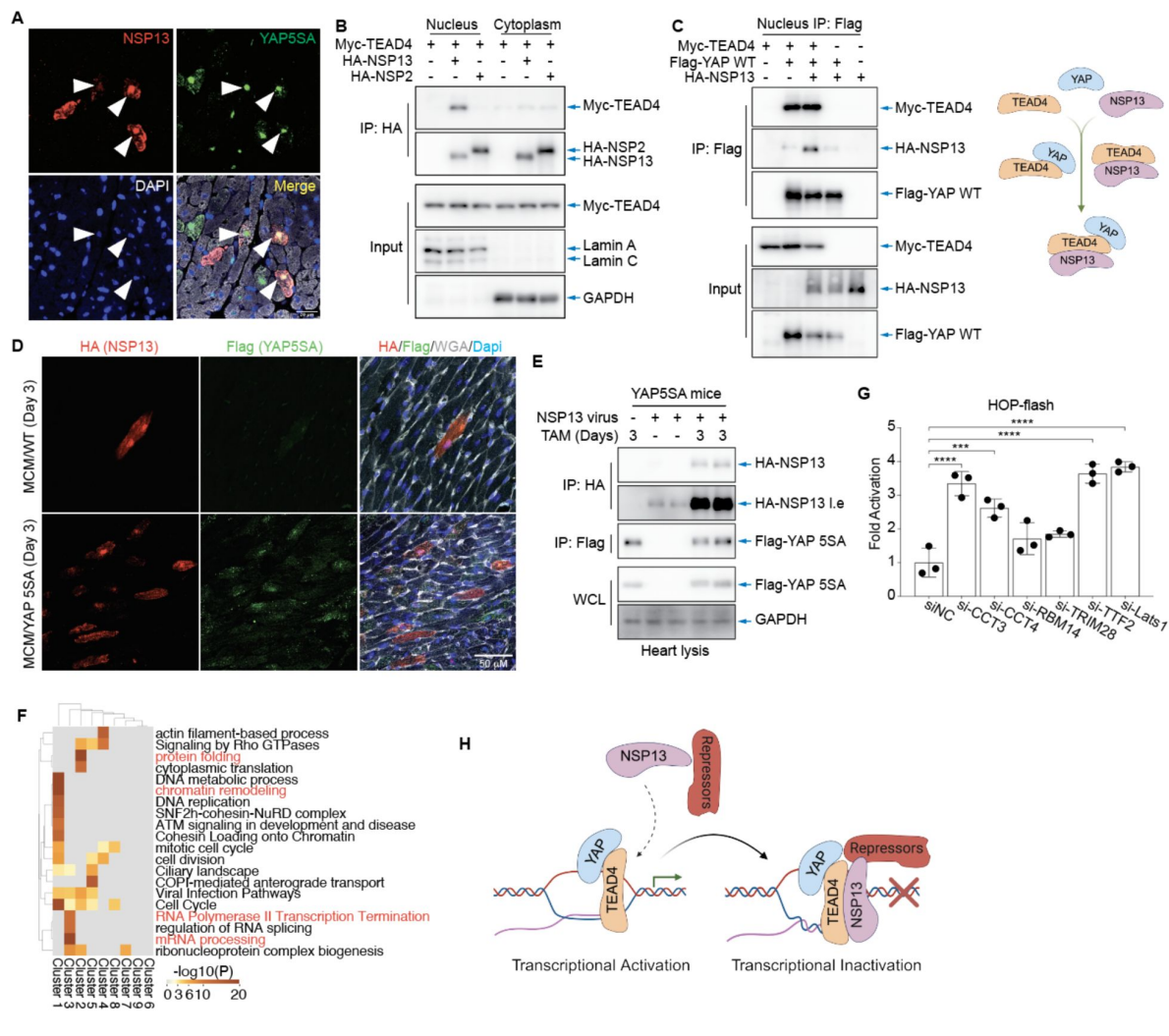
We then studied the molecular mechanism of how NSP13 suppresses YAP. Immunofluorescence data revealed that NSP13 localized in both the cytoplasm and nucleus, whereas most of YAP5SA colocalized with NSP13 in the nucleus 3 days after tamoxifen injection (**Figure 4A** [↗](#)). However, coimmunoprecipitation experiments suggested no direct interaction between NSP13 and YAP5SA (figure supplement 4A). Further investigation indicated that NSP13 associated with the transcription factor TEAD4 (**Figure 4B** [↗](#)), and both the N- and C-terminal ends of TEAD4 interacted with NSP13 (figure supplement 4B), suggesting that NSP13 may prevent YAP transactivation by competitively interacting with TEAD4. Unexpectedly, NSP13 had no effect on the YAP/TEAD4 association, but the interaction between YAP and NSP13 was much stronger in the presence of TEAD4, indicating that TEAD4 acts as a platform recruiting YAP and NSP13 (**Figure 4C** [↗](#)). Moreover, NSP13 protein levels accumulated with YAP5SA expression in cardiomyocytes (**Figure 4D, E** [↗](#)), which suggests that nuclear NSP13/YAP/TEAD4 may prevent NSP13 degradation.

Because NSP13 formed a complex with YAP/TEAD4, we hypothesized that NSP13 recruits repressors to inhibit YAP transactivation in the nucleus. The immunoprecipitation-mass spectrometry was performed to examine NSP13-interacting proteins in the presence or absence of YAP expression in HEK293T cell nuclei. We detected hundreds of candidate proteins that interact with NSP13 in the nucleus (figure supplement 4C), which were further analyzed by STRING (figure supplement 4D). Gene Ontology (GO) analysis indicated that the largest clusters are involved in RNA polymerase II transcription termination, chromatin remodeling, and protein folding (**Figure 4F** [↗](#)). We evaluated the function of a select group of these proteins in YAP transactivation by siRNA knockdown (figure supplement 4E). Reporter assay results revealed that the siRNA knockdown of CCT3 and TTF2 increased endogenous YAP transactivation in HeLa cells (**Figure 4G** [↗](#) and figure supplement 4F), suggesting that NSP13 suppressed YAP activity by recruiting suppressors to the YAP/TEAD4 complex.

## Discussion

In this study, YAP transcriptional activity was reduced after SARS-CoV-2 infection in human lung and hiPSC-CMs, and SARS-CoV-2 helicase NSP13 significantly inhibited YAP activity both *in vitro* and *in vivo*. Mechanistically, NSP13 interacted with TEAD4 and formed a complex with YAP/TEAD4 in the nucleus, which further recruited suppressors such as TTF2 and CCT3 to repress YAP-TEAD transcriptional activity (**Figure 4H** [↗](#)).

TEAD transcription factors, the major partners of YAP, are the final nuclear effectors of Hippo-YAP signaling and play critical roles in cancer development ([11](#) [↗](#), [37](#) [↗](#)). In our study, NSP13 interacted with TEAD4, but did not disrupt the YAP/TEAD4 interaction, suggesting a novel regulatory mechanism for TEAD. Immunoprecipitation-mass spectrometry showed that NSP13 interacting proteins such as TTF2 and CCT3 suppressed the YAP-TEAD4 complex. TTF2, a SWI2/SNF2 family member, facilitates the removal of RNA polymerase II from the DNA template through ATP hydrolysis ([38](#) [↗](#), [39](#) [↗](#)). Importantly, termination of the transcription complex elongation by TTF2 appears to be minimally affected by template position ([39](#) [↗](#)). CCT3 is a component of the chaperonin-containing T-complex, a molecular chaperone complex that facilitates protein folding upon ATP hydrolysis. In previous studies, CCT3 positively regulated the protein stability of YAP and served as a liver cancer biomarker ([40](#) [↗](#)). However, we found that the NSP13-interacting protein CCT3 inhibited YAP transactivation, suggesting a context-dependent function of CCT3 in YAP regulation. The detailed mechanisms require further study.



**Figure 4.**

NSP13 inactivates the YAP/TEAD4 complex by recruiting YAP repressors. **(A)** Immunofluorescence imaging showing that NSP13 colocalized with YAP5SA in cardiomyocytes of YAP5SA transgenic mice 3 days after tamoxifen injection. **(B)** Co-immunoprecipitation results suggesting that NSP13 interacts with TEAD4, a major binding partner of YAP, in the nucleus. **(C)** Results of co-immunoprecipitation experiments in nucleus of HEK293T cells showing that NSP13 did not disrupt the interaction between YAP and TEAD4, whereas TEAD4 promoted the interaction between YAP and NSP13. A working model for the YAP/TEAD4/NSP13 complex is that TEAD4 acts as a platform for recruiting YAP and NSP13. **(D-E)** Immunofluorescence imaging and western blot analysis showing that NSP13 protein levels increased after YAP5SA expression in cardiomyocytes of YAP5SA transgenic mice. **(F)** GO analysis in subclusters of NSP13 interacting proteins (SAINT, AvgP > 0.6, labelled with red in Figure S4C). **(G)** Reporter assay (HOP-flash) results showed that endogenous YAP activity was increased after the siRNA-mediated knockdown of CCT3 and TTF2 in HeLa cells. \*\*\*p < 0.001, \*\*\*\*p < 0.0001, one-way ANOVA. **(H)** Working model: NSP13, together with its interacting proteins (YAP-TEAD repressors), are recruited to the YAP/TEAD complex by interacting with TEAD4, which results in YAP-TEAD inactivation.

In conclusion, our study reveals a novel function of NSP13 in suppressing the YAP/TEAD transcriptional complex. These findings advance our understanding of the underlying effect of SARS-CoV-2 on host cells. Since it directly interacts with TEAD4 and strongly suppresses YAP transactivation, NSP13 provides a new avenue for inhibiting YAP activity as a potential treatment for YAP-driven cancer.

## Materials and methods

### Mice

We used  $\alpha$ MyHC-MerCreMer mice; wild type (WT) mice and  $\alpha$ MyHC-MerCreMer; and YAP5SA mice (26 [DOI](#)). Mice had a mixed genetic background of C57BL/6 and 129SV. The AAV9 virus (a total of  $5 \times 10^{11}$  viral genomes, 120  $\mu$ l total volume) was delivered by retro-orbital injection 2 weeks before tamoxifen injection. For the survival experiment in **Figure 2** [DOI](#), two low doses of tamoxifen (10  $\mu$ g/g) were administered to 6-week-old mice by intraperitoneal injection. The mouse cardiac function was evaluated by echocardiography at day 4 and day 8 post tamoxifen injection and all the surviving mice were sacrificed at day 21. For the experiments in **Figure 4** [DOI](#), two doses of tamoxifen (**Figure 4A** [DOI](#), 10  $\mu$ g/g; **Figure 4D-E** [DOI](#), 50  $\mu$ g/g) were injected, and the mouse hearts were collected at day 3 post tamoxifen injection.

### RNA-seq analysis

For the analysis of lung samples from patients with COVID-19, snRNA sequencing data of 7 control lungs and 35 COVID-19 lungs from GSE171668 (20 [DOI](#)) and GSE171524 (21 [DOI](#)) were downloaded and analyzed using Seurat v4 software suite (41 [DOI](#)). A total of 223,106 nuclei were used in the analysis. Each sample was normalized, and batch corrected using SCTransformation. The mitochondrial percentage was used to regress out any technical variability between batches. We used harmony (42 [DOI](#)) integration to remove technical variation among the samples. The YAP score was evaluated by using 38 canonical YAP target genes. For the analysis of iPSC-CMs infected with SARS-CoV-2, bulk RNA-seq data were obtained from a study by Perez-Bermejo et al. (25 [DOI](#)). With this dataset, we re-analyzed and evaluated the average expression of 302 cardiomyocyte-specific YAP target genes (26 [DOI](#)).

### Expression plasmids

Expression plasmids encoding HA-tagged WT, mutant, or truncated NSP13 were generated by polymerase chain reaction (PCR) and subcloned into pXF4H (N-terminal HA tag) derived from pRK5 (Genetech). Myc-tagged WT or truncated TEAD4, flag-tagged WT YAP, and flag-tagged YAP5SA were generated by PCR and subcloned into a pcDNA3 backbone. HA-tagged WT LATS2, LATS2 KR, and pRL-TK\_Luc were gifts from Dr. Pinglong Xu's lab. Reporters of HOP-flash (#83467) and 8xGT10C-luciferase (#34615) were purchased from Addgene. The NSP plasmids used in YAP transactivation screening experiments were gifts from Dr. Nevan J. Krogan's lab. All plasmids were confirmed by performing DNA sequencing.

### Cell culture and transfection

HEK293T and HeLa cells were cultured in Dulbecco's Modified Eagle Medium with 10% fetal bovine serum. Lipofectamine™ 2000 (ThermoFisher) or Lipofectamine™ 3000 (ThermoFisher) reagents were used for plasmid transfection. LipofectAmine RNAiMAX (ThermoFisher) was used for siRNA transfection.

## Luciferase reporter assay

HEK293T or HeLa cells were transfected with WT YAP-or YAP5SA responsive HOP-flash plasmid or 8xGTTC-luciferase reporter, which has an open reading frame encoding firefly luciferase, along with the pRL-Luc with *Renilla* luciferase coding as the internal control for transfection and other expression vectors (NSPs), as specified. After 24 h of transfection with the indicated treatments, cells were lysed with passive lysis buffer (Promega). Luciferase assays were performed by using a dual-luciferase assay kit (Promega). Data were quantified with POLARstar Omega (BMG Labtech) and normalized to the internal *Renilla* luciferase control.

## AAV9 viral packaging

Viral vectors were used as previously described (43 [link](#)). The construct containing HA-tagged NSP13 was cloned into the pENN.AAV.cTNT, p1967-Q vector (AAV9-HA-NSP13). Empty vector– encoding green fluorescent protein was used as the control (AAV9-GFP). Both vectors were packaged into the muscle-trophic serotype AAV9 by the Intellectual and Developmental Disabilities Research Center Neuroconnectivity Core at Baylor College of Medicine. After being titered, viruses were aliquoted ( $1 \times 10^{13}$  viral genome particles per tube), immediately frozen, and stored long-term at  $-80^{\circ}\text{C}$ . Each aliquot was diluted in saline to make a 120- $\mu\text{l}$  injection solution.

## Echocardiography

Echocardiography, for cardiac function analysis, was performed on a VisualSonics Vevo 2100 system with a 550-s probe. B-mode images and M-mode images were captured on a short-axis projection. Ejection fraction, fractional shortening, diameter of diastolic left ventricle, diameter of systolic left ventricle, and end-diastolic volume were calculated by using a cardiac measurement package installed in the Vevo2100 system.

## Histology and Immunofluorescence staining

Hearts were fixed in 4% paraformaldehyde overnight at  $4^{\circ}\text{C}$ , dehydrated in serial ethanol and xylene solutions, and embedded in paraffin. For immunofluorescence staining, slides were sectioned at 7- $\mu\text{m}$  intervals. For paraffin sections, samples were deparaffinized and rehydrated, treated with 3%  $\text{H}_2\text{O}_2$  in EtOH and then with antigen retrieval solution (Vector Laboratories Inc., Burlingame, CA, USA), blocked with 10% donkey serum in phosphate-buffered saline, and incubated with primary antibodies. The antibodies used were rabbit anti-HA (#3724, Cell signaling) and rat anti-flag (NBP1-06712, Novus Bio-logicals). Immunofluorescence stained images were captured on a Zeiss LSM 780 NLO Confocal/2-photon microscope.

## Statistical Analysis

For the data in our manuscript, the specific statistical test used is presented in the figure legend. In snRNA-seq analyses, Yap score difference between AT1 from Control and COVID19 patients was identified by using Wilcoxon test. For the analysis of iPSC-CMs infected with SARS-CoV-2, the expression score of cardiomyocyte YAP targets was evaluated by Student t-test. The mice survival rate in **Figure 2D** [link](#) was analyzed by log-rank (Mantel-Cox) test. For the other cardiac data, including EF, FS and Diameter of systolic LV, the statistical significance of the observed differences was evaluated by using three-way ANOVA and the Šidák's multiple comparisons test. For all the reporter assays in cells and heart/body ratio in Figure S2E, the statistical significance of the observed differences in mean was evaluated by using a one-way or two-way ANOVA and the post hoc Tukey's multiple comparisons test. The P value less than 0.05 was considered statistically significant (\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , \*\*\*\* $p < 0.0001$ ).

## Competing interest

J.F.M is a founder and owns shares in Yap therapeutics. J.F.M. is a co-inventor on the following patents associated with this study: patent no. US20200206327A1 entitled “Hippo pathway deficiency reverses systolic heart failure post-infarction,” patent no.15/642200.PCT/US2014/ 069349 101191411 entitled “Hippo and dystrophin complex signaling in cardiomyocyte renewal,” and patent no. 15/102593.PCT/US2014/069349 9732345 entitled “Hippo and dystrophin complex signaling in cardiomyocyte renewal.”

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## Author contributions

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Supervision: JFM

Writing – original draft: FM

Writing – review & editing: FM, JFM

## Data and materials availability

All data are available in the main text or the supplementary materials.

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## Reviewer #1 (Public Review):

In the manuscript entitled "SARS-CoV-2 NSP13 interacts with TEAD to suppress Hippo-YAP signaling", Meng *et al.* report that SARS-CoV-2 infection disrupts YAP downstream gene transcription in both patient lung samples and the iPSC-cardiomyocytes. Among the tested SARS-CoV-2 proteins, the helicase nonstructural protein 13 (NSP13) was identified to target YAP transcriptional activity both in vitro and in vivo, independent of the Hippo pathway.

Mechanistically, NSP13 inhibits YAP transcriptional activity through its interaction with TEAD4 and a group of nuclear repressor proteins, a process that requires its helicase activity. Overall, this study uncovers a novel regulation of the YAP/TEAD complex by SARS-CoV-2 infection, highlighting its impact on cellular signaling events. The manuscript is well-written and easy to follow. Here are some suggestions for the authors to further improve their work.

#### Major points

(1) The authors discovered a novel regulation of the Hippo-YAP pathway by SARS-CoV-2 infection but did not address the pathological significance of this finding. It remains unclear why YAP downstream gene transcription needs to be inhibited in response to SARS-CoV-2 infection. Is this inhibition crucial for the innate immune response to SARS-CoV-2? The authors should re-analyze their snRNA-seq and bulk RNA-seq data described in Figure 1 to determine whether any of the affected YAP downstream genes are involved in this process.

(2) The authors concluded that helicase activity is required for NSP13-induced inhibition of YAP transcriptional activity based on mutation studies (Figure 3B). This finding is somewhat confusing, as K131, K345/K347, and R567 are all essential residues for NSP13 helicase activity while mutating K131 did not affect NSP13's ability to inhibit YAP (Figure 3B). Additionally, there are no data showing exactly how NSP13 inhibits the YAP/TEAD complex through its helicase function. This point was also not reflected in their proposed working model (Figure 4H).

(3) The proposed model that NSP13 binds TEAD4 to recruit repressor proteins and inhibits YAP/TEAD downstream gene transcription (Figure 4H) needs further characterization. First, it is notable that the provided NSP13 IP-MS data did not reveal any TEAD family members as binding proteins for NSP13 (Supplement Figure 4C and the tables), suggesting that NSP13 may modulate the YAP/TEAD complex through other mechanisms, possibly involving other binding proteins. Second, NSP13 is a DNA-binding protein, and its nucleic acid-binding mutant K345A/K347A failed to inhibit YAP transcriptional activity (Figure 3B). The authors should investigate whether NSP13 could bind to the TEAD binding sequence or the nearby sequence on the genome to modulate TEAD's DNA binding ability. Third, regarding the identified nuclear repressors, the authors should validate the interaction of NSP13 with the ones whose loss activates YAP transcriptional activity (Figure 4G). Lastly, why can't NSP13 bind TEAD4 in the cytoplasmic fractionation if both NSP13 and TEAD4 are detected there (Figure 3B)? This finding indicates their interaction is not a direct protein-protein interaction but is mediated by something in the nucleus, such as genomic DNA.

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#### Reviewer #2 (Public Review):

##### Summary:

The manuscript by Meng et al. describes a potential role for the coronavirus helicase NSP13 in the regulation of YAP-TEAD activity. The authors present data that NSP13 expression in cells reduces YAP-induced TEAD luciferase reporter activity and that NSP13 transduction in cardiomyocytes blocks hyperactive YAP-mutant phenotypes in vivo. Mechanisms by which viral proteins (particularly those from coronavirus) intersect with cellular signaling events is an important research topic, and the intersection of NSP13 with YAP-TEAD transcriptional activity (independent of upstream Hippo pathway mediated signals) offers new knowledge that is of interest to a broad range of researchers.

##### Strengths:

The manuscript presents convincing data mapping the effects of NSP13 on YAP-TEAD reporter activity in the helicase domain. Moreover, the *in vivo* data demonstrating that NSP13 expression in YAP5SA mouse cardiomyocytes increased survival animal rates, and restored cardiac function is striking and is supportive of the model presented.

#### Weaknesses:

Limitations to the study are the reliance on TEAD-reporter assays to show specific effects of NPS13 on YAP-TEAD activity, incomplete characterization of the interesting *in vivo* findings that are presented, and a lack of follow-up to the proposed mechanisms identified from the IP-MS experiments.

#### Specific comments and suggestions for improvement of the manuscript:

- (1) NSP13 has been reported to block, in a helicase-dependent manner, episomal DNA transcription (PMID: 37347173), raising questions about the effects observed on the data shown from the HOP-Flash and 8xGTIC assays. It would be valuable to demonstrate the specificity of the proposed effect of NSP13 on TEAD activation by YAP (versus broad effects on reporter assays) and also to show that NSP13 reduces the function of endogenous YAP-TEAD transcriptional activity (i.e., does ectopic NSP13 expression reduce the expression of YAP induced TEAD target genes in cells).
- (2) While the IP-MS experiment may have revealed new regulators of TEAD activity, the data presented are preliminary and inconclusive. No interactions are validated and beyond slight changes in TEAD reporter activity following knockdown, no direct links to YAP-TEAD are demonstrated, and no link to NPS13 was shown. Also, no details are provided about the methods used for the IP-MS experiment, raising some concerns about potential false positive associations within the data.

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#### Author response:

##### **Reviewer #1 (Recommendations For The Authors):**

(1) Figure 3B was not cited in the manuscript.

We have now included the citation for Figure 3B in the main text: “...whereas NSP13-R567A (lost ATP consumption) and NSP13-K345A/K347A (obstructed the nucleic acid binding channel) failed to inhibit YAP activity (Figure 3B).” (Please see the revised manuscript)

##### **Reviewer #2 (Recommendations For The Authors):**

(2) In Figure 1, ciliated cells are marked as a separate cluster from “epithelial cells”. Since ciliated cells are epithelial cells, I suggest changing the nomenclature of the clusters.

We have updated the label from “Ciliated” to “Ciliated Epithelial” in Figure 1A, as suggested. (Please see the revised manuscript)

(3) Outlines of planned revisions: 1) Reanalyze snRNA-seq and bulk RNA-seq data from Figure 1 to investigate YAP target genes related to innate immune response; 2) Employ ChIP-seq to determine whether NSP13 WT or mutants (K131, K345/K347, and R567) prevent YAP/TEAD complex from binding to DNA by occupying the TEAD DNA binding site, providing insights into the mechanism; 3) Validate NSP13 interacting proteins using Immunoprecipitation-Western Blot (IP-WB) assays based on mass spectrum results; 4)

*Perform bulk RNA sequencing in cells with or without NSP13 expression to assess endogenous YAP target genes expression.*

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